

High-Level System Architecture

```
flowchart TB
    subgraph ClientApps [Client Apps]
        Client["Web Browser / Mobile<br>HTML, CSS, JS"]
    end

    subgraph BackendInfrastructure [Backend Infrastructure]
        Python["Python Detection Service<br>FastAPI"]
        Go["Go Backend API<br>Fiber + WebSockets"]
    end

    subgraph ExternalServices [External Services]
        Firebase["Firebase Auth"]
        Solana["Solana Blockchain"]
        LiveKit["LiveKit Cloud"]
        LLM["LLM Vision API<br>OpenAI/Gemini/Groq"]
    end

    Client -- "1. Authenticates" --> Firebase
    Firebase -- "Token" --> Client
    Client -- "2. API Calls & WebSockets<br>Token in Header" --> Go
    Go -- "Generates Token" --> LiveKit
    Client -- "3. Requests Stream Token" --> Go
    Client -- "4. Connects via WebRTC" --> LiveKit
    Client -- "5. Uploads Match Result" --> Python
    Python -- "6. Forwards Image" --> Go
    Go -- "Result" --> Client
```

Python -- "7. OCR Analysis" --> LLM

LLM -- "Structured Output" --> Python

Go -- "8. Executes Smart Contract" --> Solana

%% Styling

```
classDef default fill:#1e1e1e,stroke:#555,stroke-  
width:1px,color:#ddd;
```

```
classDef subgraphStyle fill:#161b22,stroke:#444,stroke-  
width:1px,color:#ddd;
```

```
style ClientApps fill:transparent,stroke:#777,stroke-  
width:1px,color:#ddd
```

```
style BackendInfrastructure  
fill:transparent,stroke:#777,stroke-width:1px,color:#ddd
```

```
style ExternalServices  
fill:transparent,stroke:#777,stroke-width:1px,color:#ddd
```